

# Music Era and Release-Market Classification

DeepMIR 2026 Fall — Homework 1

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Code, final checkpoints, README: <https://github.com/chijenpeng/26-fall-DeepMIR-HW1>

Ablation checkpoints (release):

<https://github.com/chijenpeng/26-fall-DeepMIR-HW1/releases/tag/checkpoints-v1>

October 5, 2026

## Experimental Setting

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## Pipeline

- 30 s clip, 24 kHz mono
- frozen encoder  $\rightarrow$  mean + std pooling over time
- StandardScaler  $\rightarrow$  L2 logistic regression
- train = fit, validation = model selection, test = prediction file only

## Hardware / software

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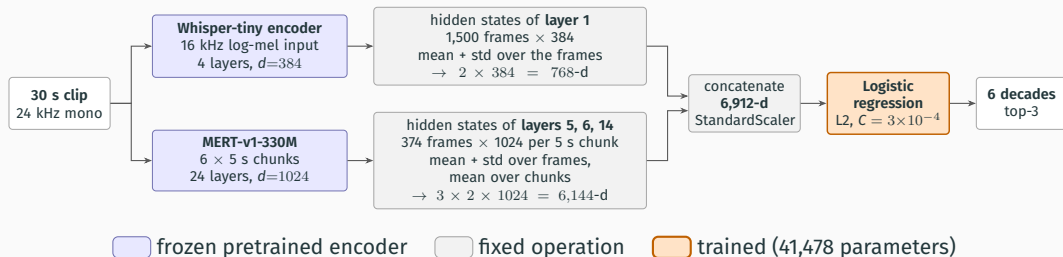
|           |                                                   |
|-----------|---------------------------------------------------|
| GPU       | 2 $\times$ NVIDIA RTX A4500 (20 GB)               |
| CPU / RAM | Intel i9-14900K, 125 GB                           |
| OS        | Ubuntu 26.04                                      |
| Software  | PyTorch 2.14, Transformers 4.45, scikit-learn 1.9 |

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## Results

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# Task 1: model



- **Whisper-tiny, layer 1:** shallow layers work best for decade; accuracy drops with depth
- **MERT-v1-330M, layers 5, 6, 14:** best three of a 25-layer sweep; adding Whisper-tiny gives +2.3 points
- **mean + std pooling:** one  $d$ -dim vector per frame  $\rightarrow$  mean and std **over the frames**  $\rightarrow 2 \times d$  values per layer
- Whisper: **1,500 frames** per 30 s clip. MERT: **374 frames** per 5 s chunk (its training length); statistics are taken per chunk, then averaged over the six
- Only the classifier is trained, with strong L2 (1,026 clips, 6,912 features)

# Task 1: release decade

**Model:** frozen encoders + logistic regression

- Whisper-tiny encoder, **layer 1**
- MERT-v1-330M, **layers 5, 6, 14**
- mean + std pooling over time (6,912-d)

Model and layers **chosen by ablation:** [▶ Section 3](#)

| Validation, 132 clips | Top-1        | Top-3        |
|-----------------------|--------------|--------------|
| Best model            | <b>52.3%</b> | <b>86.4%</b> |

## Errors

- **57%** of errors land on a **neighboring decade**
- Hardest: 2000s (recall .23), absorbed by 2010s and 1990s
- Easiest: 1960s and 2010s (one neighbor each)

| True | 1960s | 17        | 2     | 0     | 3     | 0     | 0     |
|------|-------|-----------|-------|-------|-------|-------|-------|
|      | 1970s | 5         | 9     | 3     | 2     | 2     | 1     |
|      | 1980s | 2         | 2     | 13    | 2     | 3     | 0     |
|      | 1990s | 1         | 3     | 2     | 9     | 4     | 3     |
|      | 2000s | 2         | 2     | 1     | 5     | 5     | 7     |
|      | 2010s | 0         | 2     | 0     | 0     | 4     | 16    |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

Counts. Rows = true decade (22 clips each).

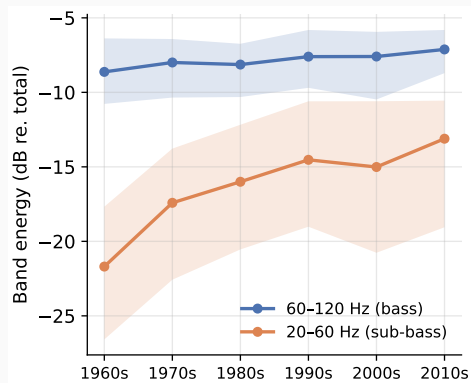
# Why neighbors get confused (1): the low end

**Hypothesis.** Old masters were cut for vinyl, where deep bass makes the stylus jump, so they should carry less sub-bass.

**Analysis.** Long-term average spectrum (Welch) of the 1,026 training clips, split into two bands.

| dB re. total | 60s   | 70s   | 80s   | 90s   | 00s   | 10s   |
|--------------|-------|-------|-------|-------|-------|-------|
| 20-60 Hz     | -21.7 | -17.4 | -16.0 | -14.5 | -15.0 | -13.1 |
| 60-120 Hz    | -8.6  | -8.0  | -8.1  | -7.6  | -7.6  | -7.1  |

- Sub-bass rises by 8.6 dB from the 1960s to the 2010s; the 60-120 Hz band barely moves
- From the 1990s on the low end is nearly identical, so it cannot separate 1990s / 2000s / 2010s



Median and inter-quartile range, training split.

## Why neighbors get confused (2): the loudness war

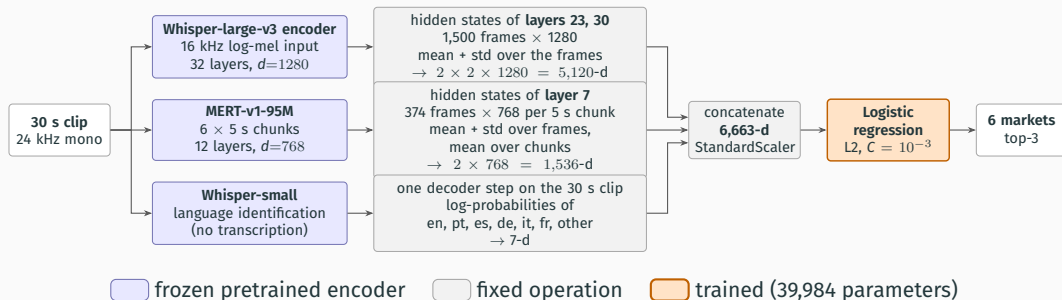
**Hypothesis.** From the 2000s, brick-wall limiters make masters louder and flatten peaks.

| Median per decade, training clips        | 60s   | 70s   | 80s   | 90s   | 00s   | 10s   |
|------------------------------------------|-------|-------|-------|-------|-------|-------|
| RMS level (dBFS)                         | -14.0 | -14.6 | -14.5 | -13.0 | -11.0 | -10.6 |
| Crest factor, peak – RMS (dB)            | 13.9  | 14.5  | 14.5  | 13.0  | 11.0  | 10.6  |
| Dynamic range, loudest 20% blocks (dB)   | 10.6  | 11.2  | 11.3  | 10.7  | 8.9   | 8.7   |
| Longest run pinned at the peak (samples) | 5     | 5     | 6     | 9     | 22    | 26    |
| Samples within 0.5 dB of the peak (%)    | 0.02  | 0.01  | 0.02  | 0.06  | 0.42  | 0.50  |

- 2000s and 2010s clips are 3 dB louder, have 3 dB less crest, and hold peaks at the ceiling. The two are **almost indistinguishable**, matching the 2000s ↔ 2010s confusion
- Production cues only split the data into “old” and “new”; finer distinctions must come from musical content

**Caveat.** The clips are 24 kHz mono, decoded from lossy Opus (YouTube). Resampling, downmixing and lossy coding reshape waveform peaks, so these numbers are trends (ANOVA  $p < 10^{-30}$ ), not exact properties of the original masters. The two sample-level rows are the most affected.

## Task 2: model



- **Whisper-large-v3, layers 23, 30**: deepest layers work best, the opposite of Task 1; this fits language being the main cue
- **MERT-v1-95M, layer 7**: +3 points on top of Whisper and language ID (57.8%  $\rightarrow$  60.8%); a music encoder adds what a speech encoder misses
- **Whisper-small language ID**: an explicit 7-d posterior; a small gain (top-3 89.2%  $\rightarrow$  90.2%) because deep Whisper layers already encode language
- Pooling as in Task 1: 1,500 Whisper frames per clip; 374 MERT frames per 5 s chunk

## Task 2: release market

**Model:** frozen encoders + logistic regression

- Whisper-large-v3 encoder, **layers 23, 30**
- MERT-v1-95M, **layer 7**
- Whisper-small language-ID posteriors

Model and layers **chosen by ablation:** [▶ Section 3](#)

| Validation, 102 clips | Top-1        | Top-3        |
|-----------------------|--------------|--------------|
| Best model            | <b>60.8%</b> | <b>90.2%</b> |

### Errors

- **US ↔ UK** is the largest confusion (9 clips)
- **UK is over-predicted:** 23 called UK, only 9 correct
- Brazil (.88) and Spain (.82) are rarely missed
- Germany, Italy: 12 of their 21 errors go to US / UK

| True | US      | 11        | 4  | 0      | 0     | 1       | 1     |
|------|---------|-----------|----|--------|-------|---------|-------|
|      | UK      | 5         | 9  | 0      | 0     | 2       | 1     |
|      | Brazil  | 0         | 0  | 15     | 0     | 0       | 2     |
|      | Spain   | 0         | 2  | 0      | 14    | 1       | 0     |
|      | Germany | 3         | 4  | 1      | 2     | 6       | 1     |
|      | Italy   | 1         | 4  | 0      | 2     | 3       | 7     |
|      |         | US        | UK | Brazil | Spain | Germany | Italy |
|      |         | Predicted |    |        |       |         |       |

Counts. Rows = true market (17 clips each).

# Why English-language markets get confused: language

The dataset has **no language label**. We estimate it with Whisper-small **language identification** on the 798 training clips, on the mixture and on Demucs-separated vocal stems.

| Market  | Home language | Home mixture | Detected as English |        |
|---------|---------------|--------------|---------------------|--------|
|         |               |              | mixture             | vocals |
| US      | English       | 92%          | –                   | –      |
| UK      | English       | 100%         | –                   | –      |
| Brazil  | Portuguese    | 87%          | 11%                 | 12%    |
| Spain   | Spanish       | 57%          | 39%                 | 41%    |
| Germany | German        | 17%          | 83%                 | 82%    |
| Italy   | Italian       | 41%          | 55%                 | 56%    |

Share of training clips whose most likely language (among en, pt, es, de, it) is the home language or English.

- Most German releases and over half of the Italian ones are **sung in English**, so their vocals carry no market cue
- Vocal stems give the same answer: the accompaniment is not what misleads the detector
- Language separates Brazil and Spain. The English-language group has to be split by genre and production style, a much weaker cue
- A release market is where the record was issued, not the artist's nationality or language

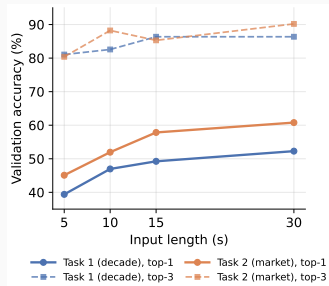
## Optional Experiments

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## Input length: 5 / 10 / 15 / 30 s

Best models on the first  $k$  seconds only; features re-extracted and classifier re-trained per length.

| Length | Task 1      |             | Task 2      |             |
|--------|-------------|-------------|-------------|-------------|
|        | Top-1       | Top-3       | Top-1       | Top-3       |
| 5 s    | 39.4        | 81.1        | 45.1        | 80.4        |
| 10 s   | 47.0        | 82.6        | 52.0        | 88.2        |
| 15 s   | 49.2        | 86.4        | 57.8        | 85.3        |
| 30 s   | <b>52.3</b> | <b>86.4</b> | <b>60.8</b> | <b>90.2</b> |



- Top-1 rises monotonically on both tasks and has **not saturated** at 30 s; Task 1 top-3 is flat after 15 s
- Task 2 gains more (+15.7 vs. +12.9 points)

## Multiple excerpts per recording

Each 30 s clip is cut into six 5 s chunks. **Chunk-level:** every chunk is its own training sample ( $6\times$  more data) and chunk probabilities are averaged at test time. **Clip-level:** chunk features are averaged first (our default).

| Task | Features                      | Clip-level |       | Chunk-level |       |
|------|-------------------------------|------------|-------|-------------|-------|
|      |                               | Top-1      | Top-3 | Top-1       | Top-3 |
| 1    | MERT-v1-95M, layers 6, 7, 9   | 47.0       | 88.6  | 51.5        | 83.3  |
| 1    | MERT-v1-330M, layers 5, 6, 14 | 50.0       | 87.9  | 50.0        | 86.4  |
| 2    | MERT-v1-95M, layer 7          | 42.2       | 72.5  | 45.1        | 76.5  |
| 2    | MERT-v1-330M, layers 8, 9, 12 | 42.2       | 69.6  | 44.1        | 76.5  |

- Task 1: no consistent gain. MERT-95M trades top-1 (+4.5 points) for top-3 (−5.3) and MERT-330M gains nothing: chunks of one recording are highly correlated
- Task 2: a small gain (+2 to +3 points top-1, +4 to +7 top-3), about 2 to 3 validation clips
- We keep clip-level features for the final models: no consistent gain, six times fewer training rows

MERT features only, not the full final models: Whisper features were extracted on whole 30 s clips.

# Task 1 as year regression

Decades are ordered, so we test a formulation that uses the order.

**Features** (same as the final Task 1 model):

Whisper-tiny encoder layer 1 + MERT-v1-330M layers 5, 6, 14, mean + std pooling, 6,912-d.

**Classification:** L2 logistic regression,  $C = 3 \times 10^{-4}$ .

**Regression:** ridge regression,  $\alpha = 10^4$ , on a decade index (0 = 1960s, 1 = 1970s, ..., 5 = 2010s), rounded to the nearest decade; top-3 = the three nearest decades.

| Formulation                       | Top-1 | Top-3 |
|-----------------------------------|-------|-------|
| Logistic regression (final model) | 52.3  | 86.4  |
| Ridge regression                  | 31.8  | 77.3  |

Mean absolute error of the regression: 0.98 decades.

| True      | 1960s | 1970s | 1980s | 1990s | 2000s | 2010s |
|-----------|-------|-------|-------|-------|-------|-------|
|           | 3     | 10    | 6     | 3     | 0     | 0     |
|           | 1     | 8     | 8     | 4     | 1     | 0     |
|           | 0     | 2     | 10    | 9     | 1     | 0     |
|           | 0     | 0     | 5     | 9     | 7     | 1     |
|           | 0     | 1     | 5     | 6     | 9     | 1     |
|           | 0     | 1     | 1     | 7     | 10    | 3     |
| Predicted |       |       |       |       |       |       |

Ridge regression. Counts; rows = true decade (22 clips each).

## Task 1 as year regression: why it fails

Both rows use the final Task 1 features (Whisper-tiny layer 1 + MERT-v1-330M layers 5, 6, 14); only the predictor differs.

Number of validation clips each formulation assigns to each decade:

| Clips predicted as                | 1960s | 1970s | 1980s | 1990s | 2000s | 2010s |
|-----------------------------------|-------|-------|-------|-------|-------|-------|
| True count                        | 22    | 22    | 22    | 22    | 22    | 22    |
| Logistic regression (final model) | 27    | 20    | 19    | 21    | 18    | 27    |
| Ridge regression                  | 4     | 22    | 35    | 38    | 28    | 5     |

- **Regression is much worse** (−20.5 points top-1): predictions collapse toward the middle, only 4 clips are called 1960s and 5 are called 2010s
- Tuning does not fix it: with a ridge penalty from  $10^2$  to  $10^6$  the best top-1 is 32%; stronger regularization worsens the collapse
- The classifier already learns the order by itself (see the t-SNE slide), so nothing is gained from imposing it

## Task 2: mixture vs. vocals vs. accompaniment

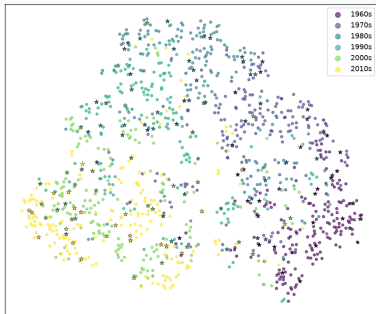
Demucs (htdemucs) separates each clip into vocals and accompaniment. The **full final pipeline** (Whisper-large-v3 layers 23, 30 + MERT-v1-95M layer 7 + language ID) is applied to each input.

| Input                            | Top-1       | Top-3       |
|----------------------------------|-------------|-------------|
| Mixture (final model)            | 60.8        | <b>90.2</b> |
| Vocals only                      | 60.8        | 87.3        |
| Accompaniment only               | 35.3        | 70.6        |
| Mixture + vocals                 | 60.8        | <b>90.2</b> |
| Mixture + vocals + accompaniment | <b>61.8</b> | <b>90.2</b> |

| Recall        | US  | UK  | BR  | ES         | DE  | IT  |
|---------------|-----|-----|-----|------------|-----|-----|
| Mixture       | .65 | .53 | .88 | .82        | .35 | .41 |
| Vocals        | .59 | .76 | .82 | .76        | .35 | .35 |
| Accompaniment | .41 | .59 | .59 | <b>.06</b> | .18 | .29 |

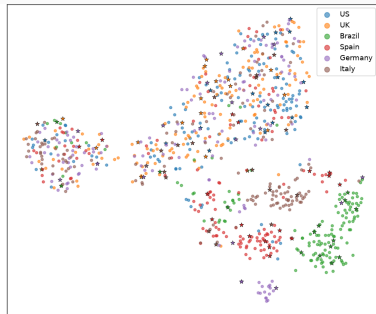
- **Market information is in the vocals:** vocals alone match the mixture; removing them costs 25 points
- Without vocals Spain falls to 6% recall, and language ID detects no Portuguese, Spanish, German or Italian at all
- Accompaniment alone stays above chance (16.7%): a weak market signal in the instrumental
- Separation does not help: all three inputs gain one clip at three times the feature size, and US ↔ UK confusions stay at 8 to 9. The final model uses the mixture

# t-SNE of the best models



**Task 1:** 6-d logit space of the classifier

- Task 1: decades line up along **one arc from the 1960s to the 2010s**, although the classifier was never told they are ordered; 2000s and 2010s overlap
- Task 2: Brazil, Spain and Italy form clusters **before any training**; US, UK and most of Germany share one English-language cloud



**Task 2:** raw feature space, no classifier

Dots = train, stars = validation. Train points in the logit space are fitted; only the stars show generalization.

# Confusion patterns and dataset bias

The detailed analyses are in the Results section. Summary, plus two additional findings:

|                        | Task 1: decade                                                                   | Task 2: market                                                   |
|------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------|
| Confusion pattern      | 57% of errors on a neighboring decade; 2000s ↔ 2010s is the worst pair           | 75% of errors among US, UK, Germany, Italy; UK is over-predicted |
| Bias in the data       | production changes with the era: sub-bass +8.6 dB, crest factor −3 dB after 2000 | English-language releases: 83% of Germany, 55% of Italy          |
| What the model keys on | mastering loudness and bandwidth as well as musical content                      | the language of the vocals rather than the market itself         |

- **Hand-crafted production features** alone reach 40.9% on Task 1 but only **22.5% on Task 2**, which is not significantly above chance (16.7%). Setting and details: next two slides
- **High-frequency energy** (8–12 kHz) is 4 to 5 dB lower in the 1960s and 1970s than later; the clips are band-limited to 12 kHz, so anything above that is unavailable

## Dataset-bias test (1): can mastering alone predict the label?

**Idea.** Describe each clip only by *how it was mastered*, never by what is played, and see how far a classifier gets. If it works, the label is partly a mastering label: a bias in the data that any model can exploit.

**Features:** 24 numbers per 30 s clip, computed from the waveform. No pretrained model.

|                         |                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spectral balance (13)   | long-term spectrum (Welch, 4096-point) $\rightarrow$ energy of 8 bands (20–60, 60–120, 120–250, 250–500 Hz, 0.5–2, 2–4, 4–8, 8–12 kHz) in dB relative to the total; below-100 Hz ratio; bass / sub-bass tilt; spectral centroid; 85% and 99% roll-off                               |
| Level and dynamics (11) | RMS level; crest factor; waveform kurtosis; share of samples within 0.5 dB of the peak; longest run of samples at the peak; median crest of 400 ms blocks; dynamic range of the loudest 20% of blocks; loudness range; block-loudness std; crest of the 60–250 Hz and 2–8 kHz bands |

**Classifier:** StandardScaler  $\rightarrow$  RBF SVM (seeded),  $C \in \{0.3, 1, 3, 10\}$  chosen by validation top-1. Trained on the training split and evaluated on validation, as in every other experiment.

## Dataset-bias test (2): result

| Validation     | Task 1      | Task 2       |
|----------------|-------------|--------------|
| Top-1          | <b>40.9</b> | 22.5         |
| Top-3          | 84.8        | 57.8         |
| Correct clips  | 54 / 132    | 23 / 102     |
| Chance top-1   | 16.7        | 16.7         |
| $p$ vs. chance | $< 0.0001$  | <b>0.076</b> |
| Final model    | 52.3        | 60.8         |

$p$ : one-sided binomial test against chance (1/6).

- **Task 1:** mastering alone reaches 78% of the final model's top-1 and separates “old” (1960s–80s) from “new” (1990s–2010s) with 73.5% accuracy: the decade label is partly a **mastering label**
- **Task 2:** not distinguishable from chance. Predictions collapse onto UK and Italy (65 of 102 clips); no German clip is recognized
- No market-specific mastering signature shows up within the 1980s, so the market signal must come from content: language and genre

|      |       |           |       |       |       |       |       |
|------|-------|-----------|-------|-------|-------|-------|-------|
| True | 1960s | 10        | 5     | 4     | 3     | 0     | 0     |
|      | 1970s | 5         | 10    | 2     | 3     | 1     | 1     |
|      | 1980s | 3         | 2     | 11    | 3     | 2     | 1     |
|      | 1990s | 0         | 6     | 3     | 5     | 7     | 1     |
|      | 2000s | 3         | 2     | 1     | 1     | 9     | 6     |
|      | 2010s | 1         | 4     | 1     | 1     | 6     | 9     |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

Task 1

|      |         |           |    |        |       |         |       |
|------|---------|-----------|----|--------|-------|---------|-------|
| True | US      | 1         | 6  | 2      | 2     | 1       | 5     |
|      | UK      | 0         | 9  | 4      | 1     | 0       | 3     |
|      | Brazil  | 0         | 5  | 6      | 1     | 0       | 5     |
|      | Spain   | 0         | 5  | 3      | 3     | 0       | 6     |
|      | Germany | 0         | 7  | 2      | 2     | 0       | 6     |
|      | Italy   | 2         | 4  | 4      | 3     | 0       | 4     |
|      |         | US        | UK | Brazil | Spain | Germany | Italy |
|      |         | Predicted |    |        |       |         |       |

Task 2

# Ablation

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## Ablation (1a): which features? Task 1

| Task 1, validation (132 clips)          | Top-1       | Top-3       |
|-----------------------------------------|-------------|-------------|
| Hand-crafted production (24-d)          | 40.9        | 84.8        |
| Whisper-tiny, layer 1                   | 47.0        | 81.8        |
| MERT-v1-95M, best layer (7)             | 47.7        | 82.6        |
| MERT-v1-95M, layers 6, 7, 9             | 47.0        | <b>88.6</b> |
| MERT-v1-330M, best layer (5)            | 51.5        | 87.9        |
| MERT-v1-330M, layers 5, 6, 14           | 50.0        | 87.9        |
| <hr/>                                   |             |             |
| Whisper-tiny layer 1 + MERT-v1-330M ... |             |             |
| layer 5 (not used)                      | 50.8        | <b>88.6</b> |
| layers 5, 6, 14 ( <b>final</b> )        | <b>52.3</b> | 86.4        |
| + hand-crafted                          | 51.5        | 86.4        |

Logistic-regression rows: mean + std pooling, one grid  $C \in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$  chosen on validation. Hand-crafted row: seeded RBF SVM.

- Music encoder > speech encoder, and a larger MERT > a smaller one
- The two families are complementary (+2.3 points from Whisper-tiny); hand-crafted features add nothing on top of MERT

### Why layers 5, 6, 14?

- **Rule, same for both MERTs:** the three best single layers of the layer sweep (95M: 7, 9, 6; 330M: 5, 14, 6)
- Alone, three layers do not beat the best single layer in top-1 (47.0 vs. 47.7, 50.0 vs. 51.5). Combined with Whisper-tiny they do: **52.3% vs. 50.8%**
- These differences are only 1 to 2 clips out of 132

## Ablation (1b): which features? Task 2

| Task 2, validation (102 clips)    | Top-1 | Top-3 |
|-----------------------------------|-------|-------|
| Hand-crafted production (24-d)    | 22.5  | 57.8  |
| MERT-v1-95M, layer 7              | 42.2  | 72.5  |
| MERT-v1-330M, layer 9             | 45.1  | 72.5  |
| Whisper-small language ID (7-d)   | 53.9  | 79.4  |
| Whisper-large-v3, layer 30        | 58.8  | 80.4  |
| + MERT-v1-330M layer 9 (not used) | 59.8  | 88.2  |
| + MERT-v1-95M layer 7             | 59.8  | 89.2  |
| + language ID                     | 59.8  | 90.2  |
| + layer 23 (final)                | 60.8  | 90.2  |

Same protocol and C grid as for Task 1.

- The order is reversed: speech encoder > explicit language ID > music encoder
- Production features are close to chance (16.7%)
- MERT adds +1 point of top-1 but **+9 points of top-3** to Whisper: a music encoder supplies what a speech encoder misses

### Why MERT-95M and not MERT-330M?

- Alone, 330M is better (45.1% vs. 42.2%)
- Added to Whisper, both give the **same top-1** (59.8%): the advantage of 330M does not carry over
- 95M has the better top-3 (89.2% vs. 88.2%) and is 3.5× smaller, so we use it. More combinations on the next slide

## Ablation (1c): Task 2 design choices

### Which MERT to combine with? (top-1 / top-3)

|                                  | 95M, layer 7 | 330M, layer 9 |
|----------------------------------|--------------|---------------|
| MERT alone                       | 42.2 / 72.5  | 45.1 / 72.5   |
| + language ID                    | 55.9 / 77.5  | 52.0 / 78.4   |
| + Whisper (30)                   | 59.8 / 89.2  | 59.8 / 88.2   |
| + Whisper (23, 30) + language ID | 60.8 / 90.2  | 59.8 / 88.2   |
| + Whisper (20, 30) + language ID | 63.7 / 91.2  | 60.8 / 90.2   |

Whisper = Whisper-large-v3 layers.

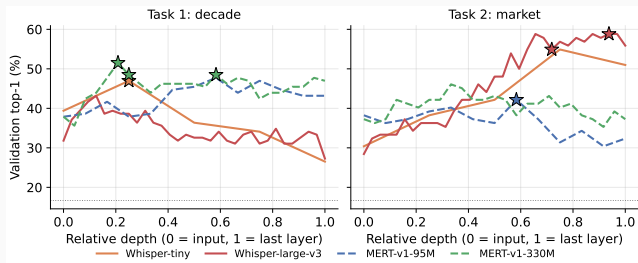
### Which second Whisper layer?

| Paired with layer 30        | Top-1 | Top-3 |
|-----------------------------|-------|-------|
| none                        | 59.8  | 90.2  |
| layer 20                    | 63.7  | 91.2  |
| layer 21                    | 62.7  | 91.2  |
| layer 22                    | 61.8  | 91.2  |
| <b>layer 23 (submitted)</b> | 60.8  | 90.2  |
| layer 27                    | 63.7  | 87.3  |
| layer 28                    | 62.7  | 88.2  |

All rows: + MERT-95M layer 7 + language ID.

- **MERT-95M, not 330M:** the larger model is better alone (+2.9 points) but equal or worse in every combination: its advantage does not carry over. In the final model MERT only adds what Whisper lacks; 330M adds 512 more dimensions for 798 training clips
- **Layer 23** was picked for its single-layer top-3 (89.2%, the best) before pairs were swept; layers 20 to 22 are 1 to 3 clips better. We keep the submitted model: this is within the noise of 102 validation clips, and picking the maximum would overfit validation further

## Ablation (2): which layer?



- One logistic regression per layer, same C grid as Ablation (1); stars mark the layers used in the final models
- **Task 1:** Whisper is best at its **first** layers (47.0% at layer 1 of Whisper-tiny) and falls to 27% at the last. MERT peaks before the middle (layer 5 of 24 in MERT-330M)
- **Task 2:** Whisper **improves with depth**, from 32% at layer 1 to 59% from layer 21 on in Whisper-large-v3. The star at layer 23 is not a top-1 peak: it was chosen for top-3 (Ablation 1c)
- Same encoder, opposite trends: decade cues sit in shallow layers, market cues in deep layers

## Ablation (3): encoder size, pooling and classifier

### Encoder size: top-1 of the best frozen layer

| Encoder          | Layers | Task 1 (layer)  | Task 2 (layer)   |
|------------------|--------|-----------------|------------------|
| Whisper-tiny     | 4      | <b>47.0</b> (1) | 54.9 (3)         |
| Whisper-base     | 6      | 37.9 (2)        | 52.9 (6)         |
| Whisper-small    | 12     | 41.7 (2)        | <b>58.8</b> (11) |
| Whisper-medium   | 24     | 43.2 (4)        | <b>58.8</b> (12) |
| Whisper-large-v3 | 32     | 43.2 (4)        | <b>58.8</b> (31) |
| MERT-v1-95M      | 12     | 47.7 (7)        | 42.2 (7)         |
| MERT-v1-330M     | 24     | <b>51.5</b> (5) | <b>46.1</b> (8)  |

### Pooling and classifier

| Task 1, MERT-v1-330M  | Top-1       | Top-3       |
|-----------------------|-------------|-------------|
| mean + std, log. reg. | <b>50.0</b> | <b>87.9</b> |
| mean only, log. reg.  | 46.2        | 87.1        |
| mean + std, RBF SVM   | 43.9        | 84.8        |

- Size does not help Whisper on Task 1: the **smallest** model is the best one. On Task 2 accuracy grows up to Whisper-small and then **saturates** at 58.8%. A larger MERT helps both tasks
- The std over time is worth +3.8 points; logistic regression beats an RBF SVM on these features
- Model selection:** encoder, layers, C and epoch are chosen by validation top-1. Over 100 configurations were compared on 132 and 102 clips, so validation scores are optimistic estimates of test accuracy

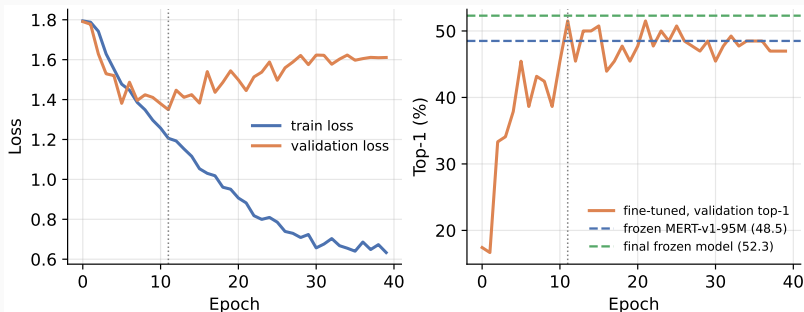
## Ablation (4): frozen features vs. fine-tuning

| Encoder                                      | Params | Epochs | Batch | Task 1 top-1            |        | Task 2 top-1            |        |
|----------------------------------------------|--------|--------|-------|-------------------------|--------|-------------------------|--------|
|                                              |        |        |       | fine-tuned (best epoch) | frozen | fine-tuned (best epoch) | frozen |
| MERT-v1-95M                                  | 95 M   | 8      | 16    | 46.2 (6)                | 47.7   | 39.2 (6)                | 42.2   |
| MERT-v1-95M                                  | 95 M   | 40     | 16    | 51.5 (11)               | 47.7   | 46.1 (16)               | 42.2   |
| MERT-v1-330M                                 | 330 M  | 40     | 16    | 48.5 (20)               | 51.5   | 44.1 (26)               | 46.1   |
| Whisper-tiny                                 | 8 M    | 20     | 8     | 34.8 (4)                | 47.0   | 50.0 (3)                | 54.9   |
| Whisper-base                                 | 20 M   | 20     | 8     | 31.8 (3)                | 37.9   | 54.9 (13)               | 52.9   |
| Whisper-small                                | 88 M   | 20     | 8     | 34.8 (4)                | 41.7   | 55.9 (2)                | 58.8   |
| Whisper-medium                               | 307 M  | 20     | 2     | 41.7 (12)               | 43.2   | 57.8 (4)                | 58.8   |
| Final frozen model (40 k trained parameters) |        |        |       |                         | 52.3   |                         | 60.8   |

**Setup.** Linear head on a learned weighted sum of all layers. AdamW, encoder lr  $2 \times 10^{-5}$  (Whisper  $10^{-5}$ ), head lr  $10^{-3}$ , one-cycle schedule, label smoothing 0.1. “Frozen” = best single layer of the same encoder + logistic regression.

- **The gain from fine-tuning is marginal.** It beats the frozen features of the same encoder in 3 of 14 runs and never beats the final model. Averaging a fine-tuned MERT with the final model adds only +0.7 (Task 1) and +2.9 points (Task 2)
- It trains 95 M to 330 M parameters instead of 40 k and is unstable **because the data are small** (1,026 and 798 training clips; next slide). **We keep the frozen model**

## Ablation (5): fine-tuning overfits



MERT-v1-95M fine-tuned on Task 1 for 40 epochs; a checkpoint was saved after every epoch.

- Validation loss is lowest at **epoch 11** and rises afterwards while training loss falls to 0.63: the model memorizes the 1,026 training clips
- The Whisper runs overfit faster (validation loss minimum at epochs 2 to 5); their learned layer weights concentrate on the *last* layer, the worst one for Task 1
- With 130 to 170 clips per class, a frozen encoder with a strongly regularized linear classifier is as accurate, more stable, and far cheaper

# Audio Language Model

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# Qwen2-Audio, zero-shot: setup

|            |                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------|
| Model      | Qwen2-Audio-7B-Instruct, no training                                                             |
| Inference  | bf16, one RTX A4500 (20 GB), batch 1, 2.2 s per clip                                             |
| Input Data | full 30 s clip, resampled to 16 kHz<br>all validation and test clips: 264 (Task 1), 204 (Task 2) |
| Decoding   | greedy, at most 16 new tokens                                                                    |

## Two outputs per clip

- **Free generation**, parsed with a regular expression (“United States” → US, etc.). No label found = *invalid*, counted as wrong
- **Log-probability ranking**: each of the six labels is scored by the log-probability of its tokens given the prompt. Every clip gets a full ranking: top-3 is defined and **no output is invalid**

**Prompt “naive”** (the audio precedes the text)

**Task 1.** *This is a 30-second excerpt of a music recording released in the United States. Our goal is to identify the decade in which the recording was released. Which decade is it? Answer with exactly one of: 1960s, 1970s, 1980s, 1990s, 2000s, 2010s.*

**Task 2.** *This is a 30-second excerpt of a music recording released in the 1980s. Our goal is to identify the country (release market) in which this record was issued. Which market is it? Answer with exactly one of: US, UK, Brazil, Spain, Germany, Italy.*

# Qwen2-Audio, zero-shot: results

| Validation, naive prompt | Task 1      | Task 2      |
|--------------------------|-------------|-------------|
| Top-1, log-prob ranking  | 34.1        | 52.9        |
| Top-3, log-prob ranking  | 69.7        | 73.5        |
| Top-1, free generation   | 31.1        | 47.1        |
| Invalid free outputs     | 1.5%        | 1.0%        |
| Our model, top-1         | <b>52.3</b> | <b>60.8</b> |
| Our model, top-3         | <b>86.4</b> | <b>90.2</b> |

|      |       |       |       |       |       |       |       |
|------|-------|-------|-------|-------|-------|-------|-------|
| True | 1960s | 14    | 8     | 0     | 0     | 0     | 0     |
|      | 1970s | 9     | 13    | 0     | 0     | 0     | 0     |
|      | 1980s | 3     | 16    | 3     | 0     | 0     | 0     |
|      | 1990s | 2     | 14    | 4     | 0     | 1     | 1     |
|      | 2000s | 0     | 14    | 1     | 0     | 0     | 7     |
|      | 2010s | 1     | 6     | 0     | 0     | 0     | 15    |
|      |       | 1960s | 1970s | 1980s | 1990s | 2000s | 2010s |

Task 1

|      |         |    |    |        |       |         |       |
|------|---------|----|----|--------|-------|---------|-------|
| True | US      | 16 | 1  | 0      | 0     | 0       | 0     |
|      | UK      | 8  | 9  | 0      | 0     | 0       | 0     |
|      | Brazil  | 1  | 1  | 15     | 0     | 0       | 0     |
|      | Spain   | 2  | 2  | 2      | 11    | 0       | 0     |
|      | Germany | 8  | 4  | 1      | 3     | 1       | 0     |
|      | Italy   | 7  | 2  | 1      | 5     | 0       | 2     |
|      |         | US | UK | Brazil | Spain | Germany | Italy |

Task 2

- Task 1: 55% of the free answers are “1970s” and none is “1990s”; the 1980s to 2000s are almost all called 1970s
- Task 2: Brazil and Spain are recognized, but English-language songs are called US: Germany 1 of 17 and Italy 2 of 17 correct
- The same weakness as our language feature: the model hears the language, not the market

Confusion matrices: naive prompt, log-probability ranking, counts, rows = true class.

## Qwen2-Audio: three prompt designs

| Task | Prompt     | Words | Log-prob ranking |             | Free generation |         |                           |
|------|------------|-------|------------------|-------------|-----------------|---------|---------------------------|
|      |            |       | Top-1            | Top-3       | Top-1           | Invalid | Most frequent free answer |
| 1    | naive      | 42    | <b>34.1</b>      | <b>69.7</b> | 31.1            | 1.5%    | 1970s, 55% of clips       |
| 1    | cues       | 74    | 26.5             | 62.1        | 24.2            | 1.5%    | 1970s, 74%                |
| 1    | production | 271   | 17.4             | 58.3        | 18.2            | 6.1%    | 2010s, 92%                |
| 2    | naive      | 43    | <b>52.9</b>      | <b>73.5</b> | 47.1            | 1.0%    | US, 54%                   |
| 2    | cues       | 78    | 22.5             | 61.8        | 28.4            | 11.8%   | US, 47%                   |

**cues:** names what to listen for (production, instrumentation; for Task 2 the language of the lyrics) and states that all classes are equally likely. **production:** a 271-word listening guide from our own analysis (sub-bass and vinyl, tape hiss, the loudness war). Full texts are in the code.

- **The longer the prompt, the worse the result:** the text prior overrides the audio. “production” ends with a description of the 2010s, and the model answers 2010s for 92% of clips
- Task 2 “cues” contains “Portuguese suggests Brazil”: in the log-prob mode 96 of 102 clips then rank Brazil first. Label words in a prompt bias token scoring
- Telling the model what to listen for does not help: the language model only receives the last Whisper layer, where decade cues are weakest (Ablation 2)

# Prompt designs, Task 1: confusion matrices

|      |       |           |       |       |       |       |       |
|------|-------|-----------|-------|-------|-------|-------|-------|
| True | 1960s | 14        | 8     | 0     | 0     | 0     | 0     |
|      | 1970s | 9         | 13    | 0     | 0     | 0     | 0     |
|      | 1980s | 3         | 16    | 3     | 0     | 0     | 0     |
|      | 1990s | 2         | 14    | 4     | 0     | 1     | 1     |
|      | 2000s | 0         | 14    | 1     | 0     | 0     | 7     |
|      | 2010s | 1         | 6     | 0     | 0     | 0     | 15    |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

**naive:** top-1 34.1%

|      |       |           |       |       |       |       |       |
|------|-------|-----------|-------|-------|-------|-------|-------|
| True | 1960s | 6         | 16    | 0     | 0     | 0     | 0     |
|      | 1970s | 4         | 18    | 0     | 0     | 0     | 0     |
|      | 1980s | 0         | 20    | 2     | 0     | 0     | 0     |
|      | 1990s | 0         | 19    | 2     | 0     | 0     | 1     |
|      | 2000s | 0         | 16    | 3     | 0     | 0     | 3     |
|      | 2010s | 0         | 13    | 0     | 0     | 0     | 9     |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

**cues:** top-1 26.5%

|      |       |           |       |       |       |       |       |
|------|-------|-----------|-------|-------|-------|-------|-------|
| True | 1960s | 1         | 0     | 0     | 0     | 0     | 21    |
|      | 1970s | 0         | 0     | 0     | 0     | 0     | 22    |
|      | 1980s | 0         | 0     | 0     | 0     | 0     | 22    |
|      | 1990s | 0         | 0     | 0     | 0     | 0     | 22    |
|      | 2000s | 0         | 0     | 0     | 0     | 0     | 22    |
|      | 2010s | 0         | 0     | 0     | 0     | 0     | 22    |
|      |       | 1960s     | 1970s | 1980s | 1990s | 2000s | 2010s |
|      |       | Predicted |       |       |       |       |       |

**production:** top-1 17.4%

- **naive:** the 1970s column absorbs the 1980s to 2000s; only the two ends (1960s, 2010s) are recognized
- **cues:** the same bias, stronger. Every decade is mostly called 1970s
- **production:** the prediction collapses onto **one column, 2010s**, the last decade described in the prompt

Validation, log-probability ranking, counts, rows = true decade (22 clips each).

## Prompt designs, Task 2: confusion matrices

| True | US      | 16        | 1  | 0      | 0     | 0       | 0     |
|------|---------|-----------|----|--------|-------|---------|-------|
|      | UK      | 8         | 9  | 0      | 0     | 0       | 0     |
|      | Brazil  | 1         | 1  | 15     | 0     | 0       | 0     |
|      | Spain   | 2         | 2  | 2      | 11    | 0       | 0     |
|      | Germany | 8         | 4  | 1      | 3     | 1       | 0     |
|      | Italy   | 7         | 2  | 1      | 5     | 0       | 2     |
|      |         | US        | UK | Brazil | Spain | Germany | Italy |
|      |         | Predicted |    |        |       |         |       |

**naive:** top-1 52.9%

| True | US      | 3         | 0  | 13     | 1     | 0       | 0     |
|------|---------|-----------|----|--------|-------|---------|-------|
|      | UK      | 2         | 0  | 13     | 2     | 0       | 0     |
|      | Brazil  | 0         | 0  | 17     | 0     | 0       | 0     |
|      | Spain   | 0         | 0  | 16     | 1     | 0       | 0     |
|      | Germany | 0         | 0  | 17     | 0     | 0       | 0     |
|      | Italy   | 0         | 0  | 16     | 1     | 0       | 0     |
|      |         | US        | UK | Brazil | Spain | Germany | Italy |
|      |         | Predicted |    |        |       |         |       |

**cues:** top-1 22.5%

- **naive:** Brazil and Spain are recognized; English-language clips from Germany and Italy go to US
- **cues:** almost every clip ranks **Brazil** first. The prompt contains “Portuguese suggests Brazil”, which raises the token score of that label
- Free generation with this prompt: US for 47% of clips, 11.8% invalid

Validation, log-probability ranking, counts, rows = true market (17 clips each).

# Disclosure

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**Tool:** Claude Code with the Claude Fable 5.1 model (Anthropic).

## Done with AI assistance

- Code implementation: feature extraction, training, evaluation, figures
- Running the experiments on the server and keeping the experiment log
- Preparing these slides

## My own

- The ideas and hypotheses: sub-bass and vinyl mastering, the loudness war and dynamics, language as the market cue
- The decisions: which experiments to run, how to read them, which models to submit, outcome interpretation

Every number in this report comes from running the code in the repository, and I reviewed the results and the text.

## References

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# References (1): pretrained models and data

## Pretrained models

- **MERT** (m-a-p/MERT-v1-95M, MERT-v1-330M): Y. Li et al., “MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training,” ICLR 2024. arXiv:2306.00107
- **Whisper** (openai/whisper-tiny, base, small, medium, large-v3): A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision,” ICML 2023. arXiv:2212.04356
- **Qwen2-Audio** (Qwen/Qwen2-Audio-7B-Instruct): Y. Chu et al., “Qwen2-Audio Technical Report,” 2024. arXiv:2407.10759
- **Demucs** (htdemucs): S. Rouard, F. Massa, A. Défossez, “Hybrid Transformers for Music Source Separation,” ICASSP 2023. arXiv:2211.08553

## Dataset

- **Discogs-VI** (source of both homework datasets): R. O. Araz, X. Serra, D. Bogdanov, “Discogs-VI: A Musical Version Identification Dataset Based on Public Editorial Metadata,” ISMIR 2024. arXiv:2410.17400

## References (2): software and methods

### Public codebases

- PyTorch: A. Paszke et al., NeurIPS 2019. torchaudio: Y.-Y. Yang et al., ICASSP 2022
- Hugging Face Transformers: T. Wolf et al., EMNLP 2020 (System Demonstrations)
- scikit-learn: F. Pedregosa et al., JMLR 2011. librosa: B. McFee et al., SciPy 2015
- nnAudio (CQT front end required by MERT): K. W. Cheuk et al., IEEE Access 2020
- Demucs code: <https://github.com/facebookresearch/demucs>
- NumPy, SciPy, pandas, matplotlib, seaborn, soundfile, tqdm, joblib

### Methods

- t-SNE: L. van der Maaten, G. Hinton, JMLR 2008. Welch spectral estimate: P. Welch, IEEE Trans. Audio Electroacoust., 1967
- Fine-tuning: AdamW (Loshchilov & Hutter, ICLR 2019), one-cycle schedule (Smith & Topin, 2019), label smoothing (Szegedy et al., CVPR 2016), SpecAugment time masking (Park et al., Interspeech 2019)

**Tool:** Claude Code, Claude Fable 5.1 (Anthropic); see the Disclosure slide.